



HABIB UNIVERSITY

Digital Logic and Design

EE-172/ CS-130/ CE-222

Spring 2025

Assignment #2

Date Released: 18th February 2025

Date Due: 25th February 2025

Max points: 100 marks

Student Name: _____

Student ID: _____

Instructors: Dr. Farhan Khan, Mr. Saad Baig, Mr. Ahmed Ali

Note: Any similarity in answers to the questions will be counted as breach of academic policy of Habib University and strict actions will be taken.

Purpose:

The purpose of this assignment is to help you understand and apply the concepts and theorems of Boolean algebra for the implementation and simplification of logic functions. In addition, this assignment will also aid you to practice the tools and techniques used to minimize logic functions.

Instructions:

1. This assignment should be done individually.
2. All questions should be answered in **black ink only**.
3. Scan your answer sheet and upload it on LMS before the due date.

Grading Criteria:

1. Your assignments will be checked by instructor/TA.
2. You can also be asked to give a viva where you will be judged whether you understood the question yourself or not. If you are unable to answer correctly to the question you have attempted right, you may lose your marks.
3. Zero will be given if the assignment is found to be plagiarized.
4. Untidy work will result in reduction of your points.

Late submission penalty:

- 1-day late submission – 20% deduction of the maximum allowable marks
- 2-days late submission – 40% deduction of the maximum allowable marks
- No submission will be accepted after two days of the original deadline

CLO Assessment:

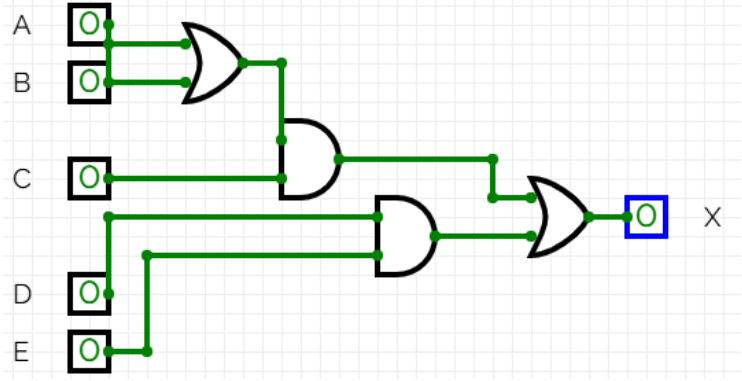
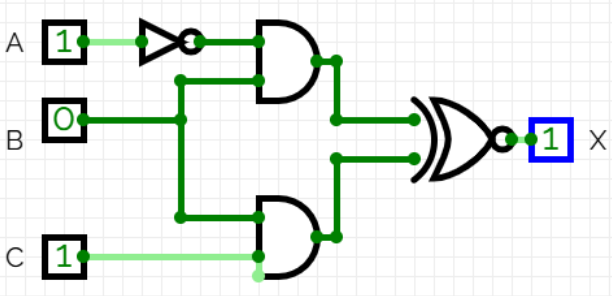
This assignment assesses students for the following course learning outcomes.

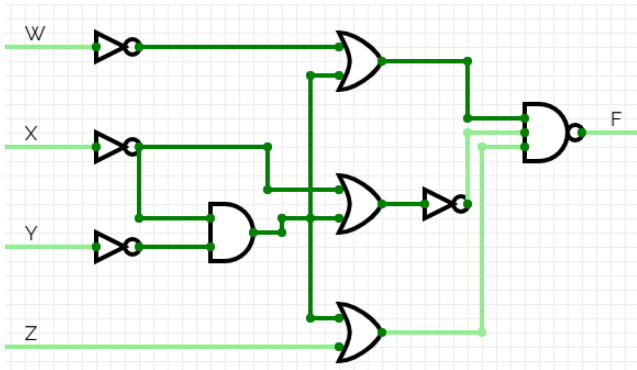
Course Learning Outcomes		CLO Assessed
CLO 1	Apply r-base (binary, octal, decimal, hexadecimal numbers systems) to digital systems and carry out arithmetic operations and conversions	

CLO 2	Apply principles of Boolean Algebra to represent and build equivalent realizations of digital logic (circuits)	✓
CLO 3	Design combinational logic circuits using logic gates	
CLO 4	Design sequential systems using finite state machine methodology	

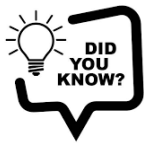
Q#	Questions	Pts
1	a) Find complement of the below expression by De-Morgan's law. Verify your solution by truth table: $F = (a + b + c + d)(a + b' + c + d)(a' + b' + c' + d')$ b) Simplify the following expressions using boolean algebra and mention the number of literals in the simplified expression. i.) $(A' + B)(A + B')(AC)$ ii.) $(z)(w + x')(wx'yz + wz)$ iii.) $(xy + z') + (xz'y + y)'$ c) We can perform logical operations on strings of bits by considering each pair of corresponding bits separately (called bitwise operation). Given two eight-bit strings A = 10001110 and B = 11001101, evaluate the eight-bit result after the following logical operations: i.) A AND B ii.) A XOR B iii.) A XNOR B	6+3x3+5
2	Simplify the following expressions using Boolean algebra postulates and theorems. Indicate all postulates or theorems used to prove the statements. i.) $xy + x(y' + z) + xyz'$ ii.) $w(x + y)(x + y')(w' + x' + z)$ iii.) $(xy + x')' + xyz' + z$	3x1
3	You're working for HU-Lock Inc. as a security systems design specialist. You receive an assignment to design a special lock system for Habib University's main gate. The client, Mr. Habootar, is quite insistent that the system must have 3 different keys to operate. However, Mr. Habootar wants the system to be quite peculiar in the following way: to open the gate, exactly two people must insert their keys simultaneously into their respective locks, and the third lock must remain empty. In the scenario where a gang of pigeons coerces the guards to unlock the gate with three keys simultaneously, an alarm is sounded. There are some additional conditions which sound the alarm listed below: C1) Key 1 inserted, Key 3 inserted → Door opens and Alarm sounds C2) Any one key inserted alone → Alarm sounds	2x2

	Let's create a truth table and Boolean expressions for this unique scenario. a) Create a single truth table for a variables Z and K. Variable Z is 1 if the door should open, and variable K is 1 if the alarm should be sounded. b) Write Minterms and Maxterms associated with the function K and express the Boolean function for only K in both canonical forms.	
4	Using De Morgan's theorems, find the complement of the following: i) $F = (x + z') + (x + y')(x + y + z')$ ii) $F = (xyz' + x'z + y + z')(x + y)$	2x2
5	Identify the form of following equation. Then simplify the following function to minimum number of literals. $F(A, B, C, D) = (A'B'D') + (A'C') + (C'D') + A'C$	2
6	For the Boolean function: $F(W, X, Y, Z) = W'X'Y'Z + XZ + W'X'YZ$ (a) Obtain the truth table of F. (b) Draw the logic diagram, using the original expression. (c) Use Boolean algebra to simplify the function, if possible, and state the minimum number of literals. (d) Draw the logic diagram from the simplified expression, and compare the total number of gates with the diagram of part (b).	2x4

7	a) Sketch NAND logic implementation:  The diagram shows a NAND logic implementation for a function of five variables A, B, C, D, and E. All inputs are 0. The circuit consists of: - A first NAND gate with inputs A and B, outputting 1. - A second NAND gate with inputs C and the output of the first gate, outputting 1. - A third NAND gate with inputs D and E, outputting 1. - A final NAND gate with inputs from the outputs of the second and third gates, outputting 0 (labeled 'X'). b) Sketch NOR logic implementation:  The diagram shows a NOR logic implementation for a function of three variables A, B, and C. Inputs are A=1, B=0, and C=1. The circuit consists of: - A first NOR gate with inputs A and B, outputting 1. - A second NOR gate with inputs B and C, outputting 1. - A final NOR gate with inputs from the outputs of the first and second gates, outputting 0 (labeled 'X').	4+2
8	Perform the following by using only Boolean algebra postulates and theorems $F(A, B, C, D) = (AD + C'B')'$ [Canonical POS] $F(w, x, y, z) = (y'z + x)(xy' + wy')(z' + yw)$ [Canonical SOP] $F(A, B, C, D) = (C + B)'(C' + D)'(BC)$ [Canonical SOP] $F'(w, x, y, z) = w'x'y + x'y'z + w'y'z$. Write F into a product of maxterm form. Note that there is complement on F in this part. - Simplify the following function, and implement it with a multi-level NAND gate circuit. $F(A, B, C, D) = (B' + C' + D')(A' + B')(A' + B')$	2x4+4
9	Simplify the following Boolean functions using K-maps: $F(w, x, y, z) = \Pi(2, 3, 7, 8, 13, 14)$ $F(A, B, C, D) = \Sigma(1, 2, 3, 6, 13, 15)$ $d(A, B, C, D) = \Sigma(9, 11, 14)$	3x2

10	You are a physicist working on a quantum computing system that stores data in memory by performing a number of Boolean operations. The storage process involves manipulating input variables through Boolean functions. For your current assignment, you need to simplify these Boolean functions for efficiency and clarity so that the quantum computer can function at full speed. Given the following Boolean functions representing parts of the storage process: $X(A, B, C, D) = \Sigma(1, 2, 3, 5, 7, 8, 10, 14)$ $Y(A, B, C, D) = \Pi(0, 1, 3, 15, 11, 12)$ a) Utilize Karnaugh Maps (K-maps) to simplify the Boolean functions X and Y, enhancing their efficiency and reducing complexity. b) Draw a logic circuit that replicates the behavior of the original Boolean functions X and Y after simplification. You can use CircuitVerse (recommended) or logic.ly and attach a properly labelled screen shot. Your goal is to streamline the encryption process while maintaining its security, so efficient Boolean function simplification and circuit design are crucial for your research.	2x5
11	You've been brought on board as a designer for HIRA to design a pigeon scaring system – when pigeons perch outside classrooms, an animatronic cat tries to scare them away. You have tasked new internees with implementing a specific logic circuit on a breadboard for testing, denoted as $F(W, X, Y, Z)$. However, there's a shortage of integrated circuits (ICs) in the test lab, making it impossible for some internees to make their circuits.  In light of this constraint, you've opted to simplify the circuit and reduce the number of gates needed. Let's break down the steps: a) To proceed, use Boolean algebra techniques to get the simplified expression of the given circuit. b) You are skeptical in your simplification and want to verify if the obtained expression is correct. For that, you make the truth table of the circuit above. c) Using the truth table, simplify the circuit using K-maps.	4x3

	d) Verify that the simplified expression obtained is the same as the one obtained in part a by drawing the simplified circuit. You may use 3 input gates if needed. Is the number of gates used after simplification the same as the one used originally?	
12	You are the chief engineer of an advanced space station orbiting a newly discovered exoplanet. The station's life support system is controlled by a digital logic circuit, but a recent analysis has revealed that unnecessary complexity in the circuit is causing excessive power consumption. To ensure the long-term sustainability of the station, you must optimize the Boolean function that governs the air circulation system. You're given the following Boolean function that represents a critical aspect of the spaceship's control system: $F(W, X, Y, Z) = \sum(1, 2, 5, 8, 10, 14)$ $d(W, X, Y, Z) = d(3, 6, 9, 11) \quad (\text{don't care conditions})$ a) Simplify the Boolean function using Karnaugh Map (K-map) reduction techniques to achieve minimal power consumption while maintaining system integrity. b) Draw a circuit diagram for the simplified design. c) Provide a breakdown of the number of terms and the number of literals in each term for both the original and reduced designs. d) Optimize for energy efficiency by redesigning the circuit in b) to use only NAND gates, which are known for their low power consumption and ease of fabrication. Provide a corresponding circuit diagram.	4+3x3



The foundational principles of Digital Logic Design are based on the mathematical work of George Boole, known as Boolean algebra, which allows complex logic operations to be broken down into simple "true" or "false" (1 or 0) binary states, making it the fundamental building block for all modern digital circuits and computers. In contrast to our common digital computers, Quantum computers use quantum bits (qubits) to store and process information, while normal computers use bits (0s and 1s). This allows quantum computers to perform certain tasks much faster than normal computers.